

Full Length Article

VPAF-FSCIL: Virtual prototype calibration and parameter-adaptive freezing for few-shot class-incremental learning

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ABSTRACT

Class Incremental Learning (CIL) enables models to learn new classes in images while retaining knowledge of previously learned classes. However, conventional CIL assumes abundant data for new classes, which is often unrealistic in real-world applications such as autonomous driving and medical diagnosis. Few-Shot Class Incremental Learning (FSCIL) addresses this issue by enabling the learning of new classes with minimal data. Nevertheless, it faces challenges like catastrophic forgetting, feature drift, and class imbalance, which can hinder effective adaptation. To tackle these challenges, we propose a method that integrates a feature-decoupled network architecture with a Parameter-Adaptive Freezing (PAF) training strategy, aiming to improve feature stability and adaptability to new classes. Inspired by Equiangular Tight Frame (ETF) theory, we decouple the model into three components: a Feature Extractor, a Feature Distributor, and an ETF Classifier. This structure integrates adaptive virtual class space allocation to ensure optimal embeddings for new classes. Moreover, PAF selectively freezes critical parameters based on the Fisher Information Matrix, which helps minimize forgetting of old classes while simultaneously improving generalization for new classes. Extensive experiments across CIFAR-100, miniImageNet, and CUB-200-2011 demonstrate that our proposed method outperforms state-of-the-art FSCIL methods in both classification accuracy and stability.

1. Introduction

Deep learning has made remarkable progress in fields such as computer vision and natural language processing, with image classification serving as a central application. By training models on large-scale datasets, they can be endowed with powerful representation capabilities (LeCun et al., 2015). Traditionally, deep learning models are trained offline under the assumption of a fixed data distribution, where all classes are learned from a static dataset at once (Szegedy et al., 2015). However, in real-world scenarios, data continuously evolves, and models are often required to incrementally learn new classes while retaining the ability to recognize previously encountered ones (De Lange et al., 2021). This capability is particularly critical in applications such as detecting novel objects in autonomous driving, identifying emerging diseases in medical diagnostics, and discovering new types of defects in industrial inspections. Consequently, Class Incremental Learning (CIL) has emerged as a significant research area. Nevertheless, CIL methods face the challenge of catastrophic forgetting (Goodfellow et al., 2013), where updates to the model's parameters during the learning of new classes can overwrite

the representations of old classes, leading to a significant drop in performance on earlier tasks.

In this work, we focus on CIL within the context of image classification. Existing CIL methods for image classification are typically divided into replay-based, regularization-based, and expansion-based strategies (Hsu et al., 2018). Although these methods have demonstrated effectiveness in mitigating forgetting, they often assume that ample data is available for new classes. However, in real-world scenarios, data for new classes is often scarce, which significantly limits the applicability and effectiveness of traditional CIL methods.

To tackle these challenges, Few-Shot Class-Incremental Learning (FSCIL) has drawn increasing attention in recent years. FSCIL aims to incrementally learn new classes from only a few labeled samples while retaining knowledge of previously learned ones. Compared to conventional CIL, FSCIL is considerably more demanding. The scarcity of data for new classes makes models highly susceptible to overfitting, while the imbalance between old and new categories further complicates training. Because the new classes contain far fewer samples, the model often struggles to maintain stable inter-class relationships, leading to weak

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generalization and limited recognition of novel categories. In addition, catastrophic forgetting remains an issue in FSCIL, as the introduction of new classes can distort the feature distributions of existing classes, leading to representation drift and performance degradation on previously learned categories.

With the growing interest in FSCIL, various strategies have been developed to improve model generalization and stability under incremental learning with limited data. One major line of work builds on meta-learning, where models are trained across multiple simulated tasks to acquire well-initialized parameters, enabling rapid adaptation to new classes with only a few examples. Another line of research focuses on feature and feature-space optimization, aiming to construct more discriminative and stable representations. Other approaches seek to enhance model robustness by incorporating principles of function optimization and representation smoothness.

Despite these advances, several challenges remain. Meta-learning approaches often depend heavily on predefined task distributions and prior information, which limits their adaptability to the continuously evolving semantic relationships among classes in real incremental scenarios. Meanwhile, feature-space optimization methods, although effective at improving inter-class separability, tend to struggle to maintain a stable embedding structure as the number of classes increases. Furthermore, many existing methods overlook the embedding drift of new classes in high-dimensional space and the resulting interference with the representations of previously learned categories, which can undermine both stability and generalization.

This paper proposes a VPAF-FSCIL method that combines a feature-decoupled network design with a parameter-adaptive freezing strategy. The core idea is to incorporate an adaptive Virtual Class Space modeling mechanism within the feature space, combined with a dynamic parameter-freezing strategy, to alleviate old-class forgetting, optimize new-class embeddings, and enhance the overall stability and adaptability of incremental learning.

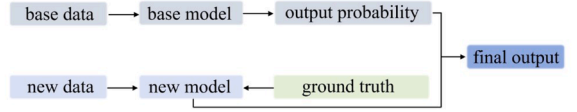
In conventional FSCIL approaches, incremental learning is typically performed directly on a standard feature extractor, which often leads to an imbalance between feature distributions of old and new classes. Such an imbalance undermines the adaptability and discrimination of newly introduced classes. To address this issue, we adopt a feature-decoupled network architecture that separates the Feature Extractor and the Feature Distributor. The Feature Distributor establishes inter-class relationships and organizes class embeddings within the Virtual Class Space (VCS) under the constraints imposed by a fixed-geometry classifier. This architectural design enables new classes to be embedded into appropriate positions in the feature space while preserving inter-class separability and maintaining geometric consistency throughout incremental sessions.

Another critical challenge in FSCIL lies in the risk that parameter updates during training may disturb the feature distributions of previously learned classes. To mitigate this issue, we introduce a Parameter-Adaptive Freezing (PAF) strategy. During the base pretraining stage, the Fisher Information Matrix (FIM) is computed to quantify the importance of each parameter, enabling the model to identify the most critical parameters for old-class representations. In subsequent incremental stages, these key parameters are selectively constrained, while less influential ones remain trainable. This mechanism alleviates catastrophic forgetting by limiting excessive drift in the feature distributions of previously learned classes. To preserve the flexibility of the classification network, a soft-freezing strategy is adopted, enabling gradual parameter adaptation and facilitating smoother integration of new classes.

By integrating the feature-decoupled architectural design with the PAF training strategy, the proposed VPAF method supports stable and efficient learning of novel classes while preserving previously acquired knowledge. A visual comparison of the overall FSCIL pipeline between existing methods and the proposed approach is illustrated in Fig. 1.

The main contributions of this work are summarized as follows:

Previous work



Our work

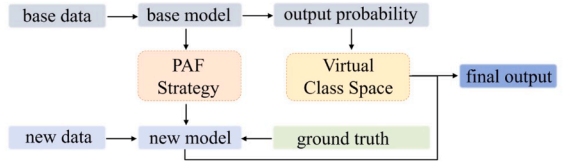


Fig. 1. Comparison of the FSCIL processing flow between previous work and our work.

1) To address the imbalance between old and new class feature distributions, we design a feature-decoupled network architecture. This design introduces a Virtual Class Space and an adaptive class space allocation mechanism within the Feature Distributor. By leveraging a well-structured geometric constraint, it enhances inter-class separability and improves feature stability during incremental learning.

2) To mitigate catastrophic forgetting caused by parameter updates, we introduce a parameter-adaptive freezing strategy. By evaluating parameter importance during the pretraining stage and selectively constraining critical parameters during incremental stages, PAF preserves the stability of old-class representations while maintaining adaptability for new ones.

3) Extensive experiments on CIFAR-100, miniImageNet, and CUB-200-2011 validate the effectiveness of the proposed design and training strategy, demonstrating reduced forgetting and improved stability during incremental learning compared with existing approaches.

2. Related works

2.1. Class incremental learning

CIL aims to build models that can continuously learn new classes without forgetting old ones. Existing research mainly focuses on three types of strategies: replay-based, regularization-based, and parameter isolation-based methods.

Replay-based methods mitigate forgetting by replaying old class samples or their approximate representations. Among them, the iCaRL (Rebuffi et al., 2017) method stores representative samples of each old class and combines nearest-neighbor classification and knowledge distillation to maintain performance on old classes. A-GEM (Chaudhry et al., 2018) constrains the current task's gradient to project within the old task's gradient space, controlling the model's update direction to avoid interference with old classes. DGR (Shin et al., 2017) uses generative adversarial networks to generate pseudo-samples for old classes, enabling replay learning without storing real data.

Regularization-based methods introduce additional loss terms to constrain the model's parameter updates, preventing drastic changes to important parameters. MAS (Aljundi et al., 2018) measures the importance of parameters in old tasks through gradient sensitivity, further guiding parameters to remain stable. LwF (Li & Hoiem, 2017) uses distillation, treating the outputs of the old model as soft targets, forcing the new model to retain the discriminative ability for old classes.

Parameter isolation-based methods explicitly allocate task-specific parameters or dynamically expand the network capacity to prevent interference between tasks. PackNet (Mallya & Lazebnik, 2018) assigns independent sub-networks to each task through pruning and

retraining. HAT (Serra et al., 2018) introduces trainable masks to achieve soft isolation between tasks. DEN (Yoon et al., 2017) achieves flexible task adaptation by using structural expansion and parameter sparsification.

Although the aforementioned methods have made significant progress in mitigating catastrophic forgetting, most of them assume that new class samples are abundant, making them unsuitable for the few-shot class-incremental learning (FSCIL) scenario, where new class data is extremely scarce. Therefore, further research focused on the FSCIL scenario has become a key direction in the field of lifelong learning.

2.2. Few-shot learning

Few-Shot Learning (FSL) aims to address the challenge of learning models with extremely limited training samples. Its core objective is to leverage prior knowledge to rapidly adapt to novel classes, and it has been widely applied in tasks such as image classification, object detection, and semantic segmentation. Existing FSL approaches can be broadly categorized into three types: meta-learning-based methods, non-meta-learning methods, and hybrid methods (Parnami & Lee, 2022). Meta-learning-based methods focus on extracting transferable learning strategies across multiple tasks to enable fast adaptation to new tasks. Representative examples include Prototypical Networks (Snell et al., 2017), which compute the mean embedding of support samples as class prototypes and classify query samples via nearest-neighbor search; and MAML (Finn et al., 2017), which learns a common initialization of model parameters that can be quickly fine-tuned with a few gradient steps when facing new tasks. Non-meta-learning methods typically rely on deep models pretrained on large-scale datasets as feature extractors, followed by simple classifier training or nearest neighbor classification. For instance, SimpleShot (Wang et al., 2019) uses pretrained features and the L2 distance for classification, achieving performance comparable to more complex meta-learning methods. Hybrid methods aim to combine multiple learning mechanisms to better handle complex real-world scenarios. AM3 (Xing et al., 2019), for example, integrates visual features and semantic label embeddings to generate more discriminative class prototypes, while semi-supervised ProtoNet (Ren et al., 2018) incorporates unlabeled data to iteratively refine prototype representations. These approaches provide both theoretical and practical support for addressing data scarcity challenges in real-world applications.

2.3. Few-shot class incremental learning

Although many CIL methods have made significant progress in mitigating catastrophic forgetting, most assume that each new class has sufficient samples, making them less suitable for real-world scenarios where data for new classes is extremely scarce. On the other hand, FSL focuses on learning new concepts from only a few labeled samples and demonstrates strong generalization, but it typically does not account for task continuity or the retention of old knowledge. To address these limitations, FSCIL has emerged as a hybrid research direction that aims to enable models to continuously learn new tasks from only a few new-class samples while still preserving recognition of previously learned classes. FSCIL faces multiple challenges, such as catastrophic forgetting, overfitting, and severe imbalance between old and new classes. Existing FSCIL methods can be broadly categorized into five groups: traditional machine learning-based methods, meta-learning-based methods, feature and feature space-based methods, replay-based methods, and dynamic structure-based methods (Tian et al., 2024).

Traditional machine learning methods typically incorporate semi-supervised learning, self-supervised learning, or statistical modeling to assist the learning process. For example, SSFSCIL (Cui et al., 2021) introduces unlabeled data in each incremental session to enhance training, while F2M (Shi et al., 2021) improves generalization on new tasks by searching for flat minima during the base training phase.

Meta-learning-based methods adapt to learning new classes incrementally by transferring knowledge from previous tasks. For example, MetaFSCIL (Chi et al., 2022) models the learning process between old and new classes as a meta-objective optimization problem, while CLOM (Zou et al., 2022) combines large-margin classification with boundary-based regulation to mitigate the performance degradation of old classes.

Feature and feature space-based methods aim to enhance model stability and generalization by optimizing feature representations or constructing well-structured feature spaces. These methods have become a central focus in current FSCIL research. FACT (Zhou et al., 2022a) proposes a forward-compatible training strategy by reserving virtual prototypes in the feature space to improve adaptability to future novel classes. LIMIT (Zhou et al., 2022b) enhances the generalization of feature representations by constructing pseudo-tasks and introduces a Transformer-based feature calibration module. C-FSCIL (Hersche et al., 2022) approaches the problem from the perspective of hyperdimensional computing, constructing approximately orthogonal class prototypes and employing a dynamically writable memory module to improve retention. NC-FSCIL (Yang et al., 2023), inspired by Neural Collapse theory, constrains old class prototypes to an ideal structure to build a more stable and discriminative classifier. PrRe (Hao et al., 2025) enhances feature space representations by introducing task and global prompts into a pre-trained ViT, combined with prototype and KNN classifiers. CoDF (Wang et al., 2024) leverages cognitive modeling by combining self-supervised learning and prior structural constraints to enhance the model's adaptability to both old and new class distributions. These methods typically involve prototype representation, subspace projection, and feature regularization, which jointly improve the retention of old knowledge and the robustness of new class learning.

Replay-based methods reproduce and integrate previously learned knowledge by storing old class samples or reconstructing their feature distributions. For example, ERDIL (Dong et al., 2021) leverages graph-based relational distillation to transfer structural information between classes, while ERDR (Liu et al., 2022) employs a generative model to synthesize pseudo-samples, thereby reducing storage requirements and mitigating privacy concerns. The Dual Distillation method (Akmel et al., 2024) leverages knowledge from base and previous-session models and incorporates semantic alignment to reduce overfitting and forgetting.

Dynamic structure-based methods adapt the model to new class learning requirements while preserving old class knowledge by dynamically expanding, pruning, or reconstructing network substructures. A representative example is TOPIC (Tao et al., 2020), which uses a neural gas network to model the topological structure among classes, while DSN (Yang et al., 2022) automatically expands or compresses network nodes in each incremental session to balance new and old knowledge.

In addition, recent studies have extended few-shot incremental learning to 3D point cloud tasks, such as COSeg (An et al., 2024b), MM-FSS (An et al., 2024a), and GFS-VL (An et al., 2025). These works share a common objective with ours—improving adaptation to novel classes while alleviating catastrophic forgetting under few-shot conditions. The key difference is that they mainly focus on 3D point cloud and multi-modal scenarios, whereas our method is tailored for FSCIL in 2D image classification, highlighting differences in data modality and technical emphasis.

In summary, although existing methods have made progress from various perspectives, enhancing the discriminability and generalization of the feature space while effectively retaining old knowledge and adapting to new classes remains a key challenge in FSCIL. Building on this insight, this paper proposes a strategy that jointly incorporates structural constraints and feature alignment to further improve the stability and effectiveness of incremental learning Fig. 2.

3. Methodology

In this section, we describe the proposed VPAF-FSCIL method in detail. First, Section 3.1 describes the problem formulation of FSCIL.

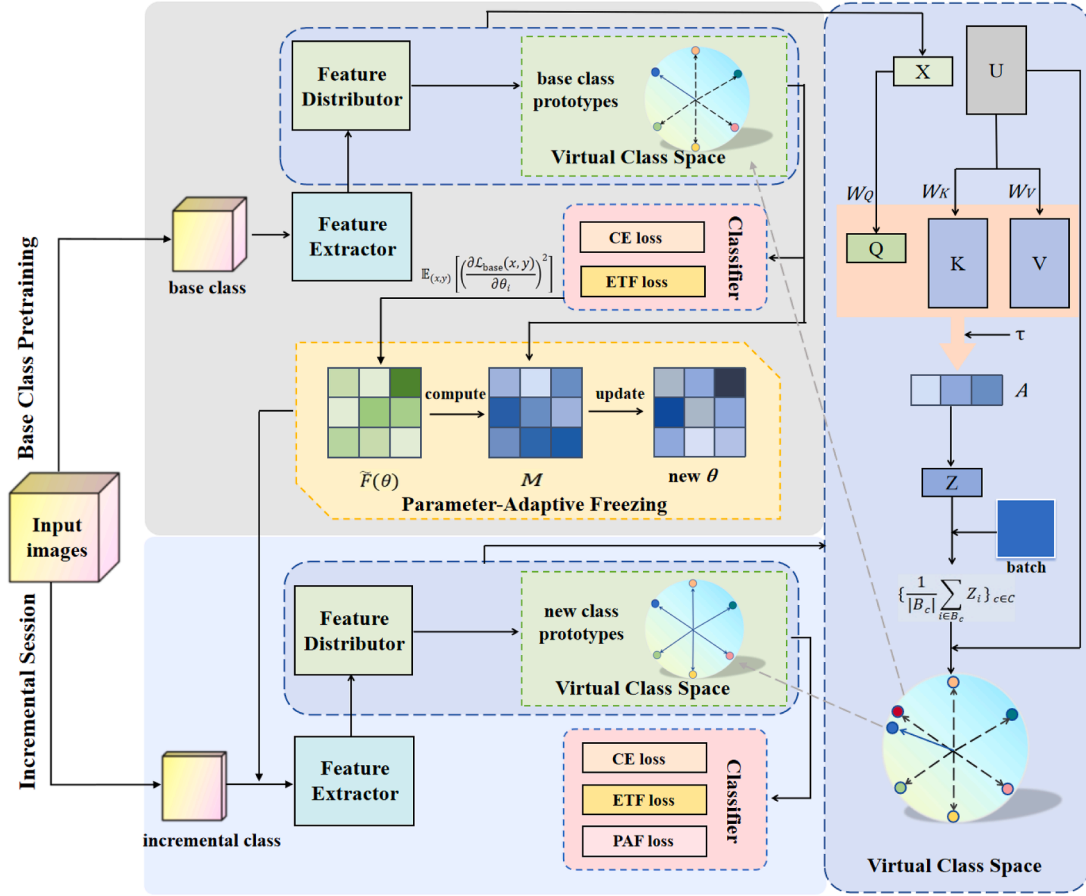


Fig. 2. Overview of the proposed VPAF-FSCIL method, which consists of two stages: base pretraining and incremental sessions. During the base pretraining stage, the Feature Extractor processes input images to obtain high-dimensional feature representations. These features are further processed by the Feature Distributor, where class prototypes are organized within the VCS under the constraints imposed by the classifier to maintain inter-class geometric structure. Meanwhile, the PAF strategy estimates parameter importance and generates a soft freezing mask to regulate subsequent parameter updates. During incremental sessions, the model is updated using both new-class samples and replay samples. New-class features are embedded into the existing VCS, and corresponding class prototypes are expanded, while the PAF strategy selectively constrains updates to parameters critical to previously learned classes, thereby supporting a balance between learning new and previously acquired knowledge.

Section 3.2 introduces the geometric motivation that guides our network design, which is derived from the Equiangular Tight Frame (ETF) structure inspired by Neural Collapse theory. In Section 3.3, we present a feature-decoupled network architecture. Then Section 3.4 introduces the PAF strategy. Finally, Section 3.5 details the implementation of the incremental training strategy.

3.1. Problem formulation

In FSCIL, the training dataset provided to the model contains base classes with abundant labeled samples and incremental classes with only a few samples in each class. The model is required to gradually learn new classes from a continuous data stream while simultaneously avoiding catastrophic forgetting of previously learned ones.

The first session is referred to as the Base Session, during which the model learns from a large number of labeled samples, enabling it to establish generalizable feature representations. The subsequent sessions are defined as Incremental Sessions, where each new class contains only a few labeled samples, making incremental learning significantly more challenging.

During each incremental session, the training data follows the N-way K-shot learning paradigm, where N denotes the number of new classes introduced and K represents the number of training samples per class. In each session, the model can access only the training data for the current

task, whereas during evaluation, it is tested on all classes encountered so far to ensure consistent performance across both old and new classes.

It is important to note that the class sets between different sessions are mutually disjoint, meaning that each session introduces a distinct set of classes that are completely independent from those in other sessions. The purpose of FSCIL is to enable the model to correctly classify all previously learned classes after completing all incremental sessions, despite the extremely limited number of samples available for each new class. The problem setup is defined as follows:

$$D_{base} = \{(x_i, y_i)\}_{i=1}^{N_{base}}, \quad y_i \in C_{base}, \quad (1)$$

where x_i denotes the sample data, y_i represents its corresponding class label, C_{base} represents the set of base classes, and N_{base} denotes the number of base class samples.

In the incremental stage t , a small number of samples from new classes are provided:

$$D_{inc}^t = \{(x_j, y_j)\}_{j=1}^{N_{inc}^t}, \quad y_j \in C_{inc}^t, \quad (2)$$

where C_{inc}^t represents the set of newly introduced classes in this stage, and $|C_{inc}^t| \ll |C_{base}|$.

To ensure class independence across sessions, we define:

$$C_{base} \cap C_{inc}^t = \emptyset, \quad C_{inc}^i \cap C_{inc}^j = \emptyset, \quad \forall i \neq j. \quad (3)$$

The target of FSCIL is that after all stages $t = 1, \dots, T$, the model can correctly classify all previously learned classes:

$$C_{total} = C_{base} \cup C_{inc}^1 \cup \dots \cup C_{inc}^T. \quad (4)$$

3.2. Theoretical background

The Equiangular Tight Frame (ETF) provides an ideal geometric formulation for class representation in the feature space. In this structure, all class centers are evenly distributed in the high-dimensional space, and the angle between any two centers is identical. Their inner product satisfies:

$$e_{k_1}^\top e_{k_2} = \begin{cases} 1, & k_1 = k_2, \\ -\frac{1}{K-1}, & k_1 \neq k_2. \end{cases} \quad (5)$$

This geometric structure ensures maximal inter-class separability and minimal redundancy.

The constant inner product $-1/(K-1)$ arises naturally from the simplex-ETF constraint, where all class centers are unit-norm vectors whose sum equals zero. This configuration enforces pairwise equal and obtuse angles between any two class centers, resulting in a uniform angular separation in the feature space. Geometrically, the negative inner product indicates that all class centers are symmetrically distributed as the vertices of a regular simplex, while the denominator $K-1$ is determined by the zero-sum and equiangular constraints of this simplex configuration. This design achieves the theoretically optimal inter-class geometry under the Neural Collapse (NC) regime, maximizing uniform angular separation and ensuring that the feature embeddings are evenly distributed on the hypersphere.

According to the Neural Collapse theory (Papayan et al., 2020), when a deep neural network is fully optimized on balanced data, the last-layer features and classifier weights naturally converge toward the ETF configuration. In this state, the within-class features collapse to their class means, all class centers are equidistant, and the classifier weights align with the corresponding class means, forming an optimal discriminative geometry.

In this work, the ETF structure is adopted to construct the feature space. The ETF constraint maintains geometric balance across classes, preventing feature drift in older classes when new classes are incrementally introduced. Meanwhile, alignment between class centers and classifier weights ensures stable, consistent decision boundaries. Such a structure renders the decision space symmetric and equidistant, thereby improving the model's robustness and interpretability during feature evolution.

3.3. Feature network design

In the FSCIL task, traditional end-to-end feature learning methods often struggle to maintain stable inter-class structural information during the incremental stages. Therefore, this study decouples the feature learning process into three key components: Feature Extractor, Feature Distributor, and Classifier. The Feature Extractor is responsible for extracting deep feature representations from input images, the Feature Distributor transforms the extracted features to regulate their relative positions in the feature space, facilitating a more stable inter-class structure during incremental learning, and the Classifier employs an Equiangular Tight Frame (ETF) structure for classification to enhance inter-class separability.

During the feature extraction phase, an input image x is processed by the feature extractor f_{θ_f} , yielding a high-dimensional feature representation:

$$X = f_{\theta_f}(x), \quad X \in \mathbb{R}^d, \quad (6)$$

where d represents the dimension of the feature vector, which is the dimensionality of each class center vector. During the base pretraining

stage, the feature extractor is optimized in an end-to-end manner. However, during the incremental stage, adaptive partial parameter freezing is applied based on the results computed by the PAF strategy to maintain the stability of old-class features.

During the feature distribution stage, the Feature Distributor g_{θ_d} further optimizes the feature representations, ensuring that new class embeddings can adapt to the existing class feature distribution.

Prototype-based FSCIL methods typically constrain the feature distribution within a fixed geometric structure—such as orthogonal or Equiangular Tight Frame (ETF) structures—which assume that class features are symmetrically and uniformly distributed. These strict constraints are effective when data are balanced and abundant but tend to fail under few-shot or highly imbalanced conditions. In such cases, limited samples of new classes may not align well with the pre-existing feature geometry, leading to concept drift and feature imbalance. When new classes are introduced, the feature centers of old classes may be perturbed, leading to inter-class imbalance and reducing the stability of new class embeddings.

To overcome this limitation, we propose a Feature Distributor built upon a Virtual Class Space (VCS). Unlike a fixed geometric structure, the VCS acts as a dynamically adjustable class embedding space that can reorganize its structure when new classes arrive. Before the introduction of new classes, a virtual space is pre-defined to represent the current inter-class relationships. When new-class data are provided, this space adjusts their embeddings by leveraging the statistical and geometric relationships among existing classes. This design enables the model to maintain a globally coherent geometric structure while providing local flexibility for new-class embeddings, thereby reducing concept drift and improving adaptability to complex, unbalanced data.

Within the Feature Distributor, we employ an attention-guided feature transformation strategy to adaptively align new class features with the existing class structure.

Given an input feature X from the feature extractor, we define the new-class query, old-class keys, and value vectors as:

$$Q = W_Q X, \quad K = W_K U, \quad V = W_V U, \quad (7)$$

where W_Q, W_K, W_V are trainable projection matrices that map the input feature and the class prototypes into the query, key, and value spaces, respectively. Q is the query vector derived from the input feature X , while K, V denote the key and value matrices of the existing class centers U stored in the virtual class space. Each row of U represents one class prototype. This design allows the attention module to adaptively align new-class features with the existing class structure.

The attention weights between the new-class feature and all existing class centers are computed as:

$$A_{i,j} = \frac{\exp\left(\frac{Q_i K_j^\top}{\sqrt{d \cdot \tau}}\right)}{\sum_{j=1}^{C_{old}} \exp\left(\frac{Q_i K_j^\top}{\sqrt{d \cdot \tau}}\right)}, \quad (8)$$

where Q_i denotes the query vector of the i th sample in the current batch. The temperature parameter τ controls the sharpness of the attention distribution and is defined as:

$$\tau = \sigma(W_\tau) + \epsilon, \quad \epsilon > 0, \quad (9)$$

where $\sigma(\cdot)$ denotes the Sigmoid function, and W_τ is a trainable parameter. The Sigmoid function ensures that τ is bounded within $(0, 1)$, and the small positive constant ϵ prevents τ from approaching zero, thus avoiding numerical instability and disappearance of gradients. Beyond positivity, τ also regulates the sharpness of the attention distribution in Eq. (8). A larger τ results in smoother and more uniform attention weights, allowing new class embeddings to attend to multiple old class centers, improving adaptability. Conversely, a smaller τ sharpens the focus on the most similar old classes, enhancing inter-class separability.

We define the attention vector A_i for the i th new class sample, which represents the similarity weights between the new class sample and all

old class centers. The vector A_i is of dimension $1 \times C_{old}$, where C_{old} is the number of old classes:

$$A_i = [A_{i,1}, A_{i,2}, \dots, A_{i,C_{old}}] \in \mathbb{R}^{1 \times C_{old}}, \quad (10)$$

where $A_{i,j}$ denotes the attention weight between sample i and the j th old class center, measuring the similarity between the new class sample and the old class centers. The attention weights are computed using the Softmax function, ensuring that each vector A_i has non-negative elements that sum to 1.

The refined embedding for each sample is obtained through a residual-based fusion between its original feature and the attention-weighted old-class centers:

$$Z_i = \frac{\tilde{Z}_i}{\|\tilde{Z}_i\|_2} = \frac{X_i + A_i V}{\|X_i + A_i V\|_2}, \quad (11)$$

where $\|\cdot\|_2$ denotes the Euclidean (L_2) norm.

The residual design enables smooth integration of new-class features into the old class manifold, while normalization ensures geometric consistency and prevents scale drift.

This process encourages new-class features to lie within the convex hull of old class centers, which maintains stability and prevents over-deviation from the established feature topology.

Finally, compute the feature center for each new class in the batch and update it into the known classes in the virtual class space:

$$U_{updated} = U \cup \left\{ \frac{1}{|B_c|} \sum_{i \in B_c} Z_i \right\}_{c \in C}, \quad (12)$$

where B_c represents the set of samples belonging to class c in the current mini-batch, and C represents the total number of classes in the current task.

Through the design of the Feature Distributor, the feature embeddings of new classes can be adaptively adjusted to better align with the existing class structure. By utilizing a convex combination approach, we ensure that the embeddings of new classes are positioned closer to the centers of similar old classes, while remaining stably within the convex hull of the virtual class centers, thereby preventing feature drift.

We initialize the output class size of the classifier based on the number of new classes introduced in the current task. Given that the virtual class space is dynamically adjusted in the Feature Distributor, we add an ETF-based classifier to further enforce the even distribution of all class centers in the feature space according to the designed geometric structure, ensuring that their positions satisfy maximal angular separation. The classifier imposes the following constraint on the class centers:

$$W_i^T W_j = -\frac{1}{C_{total} - 1}, \quad i \neq j, \quad (13)$$

where W_i and W_j represent the class centers of classes i and j .

3.4. Parameter-adaptive freezing strategy

In FSCIL, a core challenge is maintaining the stability of knowledge from previous classes while introducing new ones. Due to the scarcity of samples for new classes, gradient updates can easily perturb the key parameters of old classes, leading to catastrophic forgetting. Although existing data replay mechanisms partially mitigate this issue by reusing a few old samples, they do not explicitly regulate parameter-level interference.

To address this challenge, we propose the PAF strategy, which estimates parameter importance and applies adaptive constraints on critical parameters, thus achieving a trade-off between stability and plasticity.

The Fisher Information Matrix is a classic metric used to quantify the sensitivity of the model's output distribution $p(y|x; \theta)$ to small perturbations in parameters θ . The value $F(\theta_i)$ represents the expected curvature of the loss surface with respect to the parameter θ_i , where a higher value indicates that the parameter is more important for the model's performance.

The Fisher Information is defined as follows:

$$F(\theta_i) = \mathbb{E}_{(x,y)} \left[\left(\frac{\partial}{\partial \theta_i} \log p(y|x; \theta) \right)^2 \right]. \quad (14)$$

For classification tasks, since the cross-entropy loss corresponds to the negative log-likelihood, it can be approximated as:

$$F(\theta_i) \approx \mathbb{E}_{(x,y) \sim D_{base}} \left[\left(\frac{\partial \mathcal{L}_{base}(x, y)}{\partial \theta_i} \right)^2 \right]. \quad (15)$$

Intuitively, the Fisher Information reflects the extent to which the model's loss changes when a parameter is slightly perturbed. As such, it is well-suited for assessing parameter importance and sensitivity. This approximation effectively measures the average sensitivity of parameters to changes in the loss, providing a theoretical basis for the freezing strategy.

During the base pretraining stage, the model is optimized by:

$$\mathcal{L}_{base} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{ETF}, \quad (16)$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss and $\mathcal{L}_{ETF}$ enforces geometric regularity in the ETF classifier.

We estimate the Fisher Information based on the base dataset D_{base} . In practical training, the expected value of the Fisher Information often has good statistical properties. Therefore, we use mini-batch sampling, which ensures computational efficiency while adequately reflecting the statistical stability of the gradient directions. Empirically, mini-batch estimation is equivalent to introducing finite variance noise to the Fisher expectation. Since the subsequent Sigmoid normalization smooths out the differences, this noise has minimal impact on the final freezing mask.

$$\tilde{F}(\theta) = F(\theta) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2). \quad (17)$$

To avoid rigid hard-freezing, we define a differentiable soft-freezing mask:

$$M_i = 1 - \alpha \cdot \sigma \left(\frac{\tilde{F}(\theta_i) - s_{adaptive}}{s_{adaptive}} \right), \quad (18)$$

where α is the freezing intensity parameter, which controls the degree of freezing for important parameters and affects the strength of PAF soft freezing. When $\alpha = 0$, parameters are not frozen and update freely, while when $\alpha = 1$, they are fully frozen. $s_{adaptive}$ is the adaptive Fisher importance threshold, which is used to distinguish between "important parameters" and "non-important parameters".

Finally, the gradient update under the PAF strategy is expressed as:

$$\theta_i \leftarrow \theta_i - \eta \cdot M_i \cdot \frac{\partial \mathcal{L}_{base}}{\partial \theta_i}. \quad (19)$$

The M_i values for important parameters are small, which suppresses their updates during the incremental process. In contrast, parameters deemed less important maintain higher updatability.

This soft-freezing mechanism effectively preserves key parameters crucial to base-class knowledge while allowing less critical parameters to remain plastic.

3.5. Incremental training strategy

In the base pretraining stage, the model learns features of all base classes from the base dataset D_{base} . By optimizing the cross-entropy loss $\mathcal{L}_{CE}$ and ETF geometric constraint loss $\mathcal{L}_{ETF}$, the model effectively learns base class features and ensures geometric separation between classes.

In the incremental training stage, the model continues to learn new classes based on the knowledge gained during the base pretraining stage. During this stage, the PAF strategy is applied to ensure the stability of old class knowledge while allowing the model to flexibly learn new class features. To mitigate feature drift in old classes, the incremental training stage also employs a data replay strategy to retain old-class knowledge.

This strategy combines new class data with replayed data, ensuring that the model learns new classes while retaining prior class knowledge at each incremental stage.

After each incremental session, a subset of samples from the current session is randomly selected and stored to update the replay buffer D_{replay} , ensuring that old class knowledge is preserved for future incremental sessions:

$$D_{\text{replay}} = \{(x_i, y_i) \mid (x_i, y_i) \sim \text{Uniform}(D_{C^{t-1}})\}, \quad (20)$$

where C^{t-1} denotes the set of all classes seen up to step $t-1$, and $D_{C^{t-1}}$ denotes their corresponding training data. The notation $\sim \text{Uniform}(\cdot)$ indicates that (x_i, y_i) are randomly sampled with equal probability from the dataset of these classes.

For the current incremental session t , the new class data is loaded and combined with the data from the replay buffer D_{replay} to construct the training dataset D_{train} .

In each mini-batch, the total loss is computed, which consists of the cross-entropy loss, ETF regularization loss, and the PAF soft loss. The PAF-related regularization term suppresses large gradient updates for important parameters, preventing the disruption of old class features:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{ETF} + \lambda_2 \sum_i (1 - M_i) \left(\frac{\partial \mathcal{L}_{\text{base}}}{\partial \theta_i} \right)^2. \quad (21)$$

The complete training procedure, including both base class pre-training and incremental training, is outlined in [Algorithm 1](#).

Algorithm 1 Overall training procedure of VPAF-FSCIL.

```

1: Input: Base dataset  $D_{\text{base}}$ , Incremental datasets  $\{D_{\text{inc}}^1, D_{\text{inc}}^2, \dots, D_{\text{inc}}^T\}$ ,
   Learning rate  $\eta$ , trade-off coefficients  $\lambda_1, \lambda_2$ , Freezing parameters
    $\alpha, s_{\text{adaptive}}$ 
2: Output: Trained model parameters  $\theta$  and classification result
3: Initialize feature extractor  $f(\cdot)$ , feature distributor  $g(\cdot)$ , and classifier
    $h(\cdot)$ 
4: for each mini-batch  $(x, y) \in D_{\text{base}}$  do
5:    $X \leftarrow f(x)$  ▷ Feature extraction
6:    $Z \leftarrow g(X)$  ▷ Attention-based refinement
7:   Compute classification loss  $L_{\text{base}}$  using Eq. \(16\)
8:   Backpropagate and update  $\theta$ 
9: end for
10: Compute Fisher Information  $F(\theta_i)$  over mini-batches using Eq. \(15\)
11: Compute soft mask  $M_i$ 
12: for each incremental session  $t = 1 \dots T$  do
13:   Load  $D_{\text{inc}}^t$  and replay buffer  $D_{\text{replay}}$ 
14:   Construct  $D_{\text{train}} = D_{\text{inc}}^t \cup D_{\text{replay}}$ 
15:   for each mini-batch  $(x, y) \in D_{\text{train}}$  do
16:      $X \leftarrow f(x)$ 
17:      $Z \leftarrow g(X)$ 
18:     Compute total loss  $L$  using Eq. \(21\)
19:     Update  $\theta$  using Eq. \(19\)
20:   end for
21:   Update  $D_{\text{replay}}$  with selected samples from  $D_{\text{inc}}^t$ 
22: end for

```

4. Experiments

In this section, we evaluate our proposed method on three widely-used FSCIL benchmark datasets and compare it with state-of-the-art approaches. Additionally, we conduct ablation studies to verify the effectiveness of each proposed component.

4.1. Datasets

CUB-200-2011 is a fine-grained bird classification dataset containing 200 categories. Following the FSCIL setting, we use 100 base classes

and introduce the remaining 100 classes in 10 incremental sessions, each adding 10 classes with 5 samples per class. Images are resized to 256×256 and then center-cropped to 224×224 for training.

MiniImageNet is a subset of ImageNet with 100 classes, each having 600 images. Following prior works, we use 60 base classes and incrementally introduce 40 novel classes over 8 incremental sessions. Each session adds 5 new classes, and each new class has only 5 training samples. All images are resized to 84×84 pixels.

CIFAR-100 contains 100 categories, each with 600 RGB images of size 32×32 pixels. For FSCIL, we follow the standard setting, using 60 classes for the base training session and 40 for incremental sessions. These 40 classes are divided into 8 incremental sessions, each introducing 5 new classes with only 5 labeled training samples per class.

4.2. Implement details

All experiments are conducted on the PyTorch platform. The analysis of hyperparameters in PAF can be found in [Section 4.6](#). In the ablation experiments and comparisons with other methods, previous works commonly use ResNet12, ResNet18, and ResNet20 for experiments. We follow the standard setups of prior methods such as CEC, WaRP, and LIMIT, using ResNet20 as the feature extractor for CIFAR-100 and ResNet18 as the feature extractor for miniImageNet and CUB-200-2011. During pre-training, we use Stochastic Gradient Descent with a momentum of 0.9 and weight decay of $5e-4$.

For CIFAR-100 and miniImageNet, we train from scratch for 150 epochs with a batch size of 64 and an initial learning rate of 0.01, using cosine annealing for the learning rate decay. For CUB-200-2011, the initial learning rate is 0.01, and we use the StepLR learning rate scheduler with a step size of 25.

During incremental sessions, we adopt the proposed PAF strategy to stabilize old class knowledge and adapt to new class embeddings.

In the implementation process, the freezing intensity parameter α and the adaptive threshold s_{adaptive} are critical hyperparameters that affect both the stability of the model and its adaptability during incremental learning. The parameter α is selected via grid search on the validation set and set to 0.7 to balance retaining old classes and adapting to new ones.

4.3. Evaluation metrics

We evaluate classification performance using accuracy and quantify the forgetting phenomenon with a forgetting degree metric. As incremental learning spans multiple sessions, directly computing the difference between the final and base session accuracies fails to capture intermediate forgetting dynamics. Therefore, we compute the accuracy drop at each incremental session relative to the preceding session and define the total forgetting degree as the sum of squared accuracy declines, formulated as:

$$F = \sum_{t=1}^T \max(0, A_{t-1} - A_t)^2, \quad (22)$$

where F represents the cumulative forgetting across sessions, A_t is the accuracy in the t^{th} incremental phase, and A_{t-1} refers to the accuracy in the preceding session.

In addition to accuracy, we report the Degradation Rate (DR), which measures the relative performance drop from the base session to the final incremental session. DR quantifies how much accuracy is lost as the model progresses through incremental sessions, and is defined as:

$$DR = \frac{A_0 - A_T}{A_0} \times 100\%, \quad (23)$$

where A_0 denotes the classification accuracy after the base session, A_T denotes the classification accuracy after the final incremental session T . A lower DR value indicates better stability and less catastrophic forgetting, reflecting the model's ability to retain old knowledge while adapting to new classes.

Table 1

Classification accuracy (%) on mini-ImageNet in the 5-way 5-shot FSCIL setting. Acc (%) represents the average classification accuracy across all incremental sessions. DR (%) Represents the performance degradation rate.

Method	0	1	2	3	4	5	6	7	8	Acc	DR
CEC	72.00	66.83	62.97	59.43	56.70	53.73	51.19	49.24	47.63	57.75	33.85
WaRF	72.99	68.10	64.31	61.30	58.64	56.08	53.40	51.72	50.65	59.69	30.61
LIMIT	72.32	68.47	64.30	60.78	57.95	55.07	52.70	50.72	49.19	59.06	31.98
C-FSCIL	76.40	71.14	66.46	63.29	60.42	57.46	54.78	53.11	51.41	61.61	32.71
ALICE	80.60	70.60	67.40	64.50	62.50	60.00	57.80	56.80	55.70	63.99	30.89
SAVC	81.12	76.14	72.43	68.92	66.48	62.95	59.92	58.39	57.11	67.05	29.60
OrCo	83.27	73.94	70.51	66.37	64.19	61.14	59.19	57.37	56.34	65.81	32.34
NC-FSCIL	84.02	76.80	72.00	67.83	66.35	64.04	61.46	59.54	58.31	67.82	30.60
metaNC	85.18	80.32	76.01	72.35	69.25	66.11	63.07	61.00	59.04	70.26	30.60
VPAF (Ours)	85.35	81.39	78.25	75.40	72.22	69.46	66.95	64.21	61.82	72.78	27.57

Table 2

Classification accuracy (%) on CIFAR-100 in the 5-way 5-shot FSCIL setting.

Method	0	1	2	3	4	5	6	7	8	Acc	DR
CEC	73.07	68.88	65.26	61.19	58.09	55.57	53.22	51.34	49.14	59.53	32.75
WaRF	80.31	75.86	71.87	67.58	64.39	61.34	59.15	57.10	54.74	65.82	31.84
LIMIT	73.81	72.09	67.87	63.89	60.70	57.77	55.67	53.52	51.23	61.84	30.59
C-FSCIL	77.47	72.40	67.47	63.25	59.84	56.95	54.42	52.47	50.47	61.64	34.85
ALICE	79.00	70.50	67.10	63.40	61.20	59.20	58.10	56.30	54.10	63.21	31.52
SAVC	78.77	73.31	69.31	64.93	61.70	59.25	57.13	55.19	53.12	63.63	32.56
OrCo	79.80	63.12	62.41	59.96	58.69	56.47	55.36	53.95	50.98	60.08	36.12
NC-FSCIL	82.52	76.82	73.34	69.68	66.19	62.85	60.96	59.02	56.11	67.50	32.00
VPAF (Ours)	82.73	78.21	74.18	71.54	67.79	63.86	61.92	60.23	58.27	68.75	29.57

Table 3

Classification accuracy (%) on CUB-200 in the 10-way 5-shot FSCIL setting.

Method	0	1	2	3	4	5	6	7	8	9	10	Acc	DR
CEC	75.85	71.94	68.50	63.50	62.43	58.27	57.73	55.81	54.83	53.52	52.28	61.33	31.07
FACT	75.90	73.23	70.84	66.13	65.56	62.15	61.74	59.83	58.41	57.89	56.94	64.42	24.98
WaRF	77.74	74.15	70.82	66.90	65.01	62.64	61.40	59.86	57.95	57.77	57.01	64.66	26.67
LIMIT	76.32	74.18	72.68	69.19	68.79	65.64	63.57	62.69	61.47	60.44	58.45	66.67	23.41
ALICE	77.40	72.70	70.60	67.20	65.90	63.40	62.90	61.90	60.50	60.60	60.10	65.75	22.35
SAVC	81.85	77.92	74.95	70.21	69.96	67.02	66.16	65.30	63.84	63.15	62.50	69.35	23.64
OrCo	74.62	71.40	68.75	65.59	63.68	60.82	59.23	57.50	55.55	54.90	53.61	62.33	28.16
NC-FSCIL	80.45	75.98	72.30	70.28	68.17	65.16	64.43	63.25	60.66	60.01	59.44	67.28	26.11
VPAF (Ours)	81.98	77.94	75.18	72.26	68.59	67.42	66.29	65.96	64.24	63.41	62.64	69.63	23.59

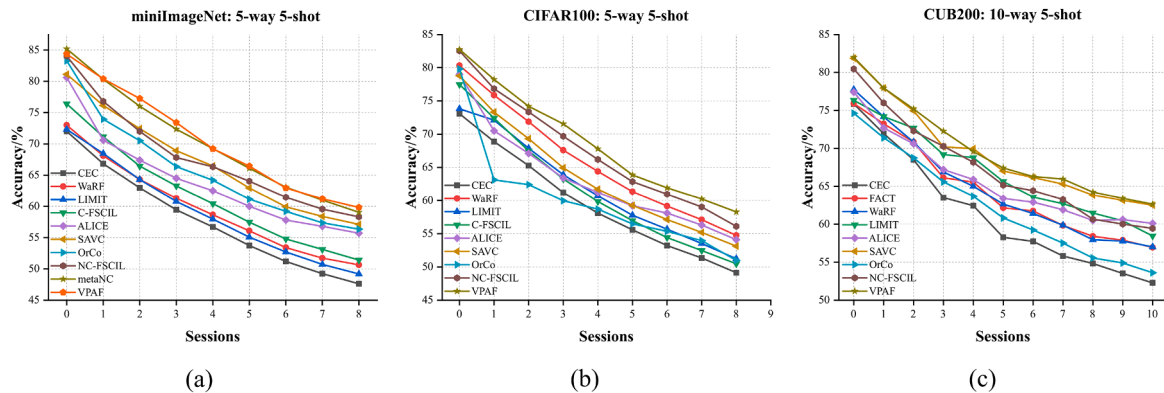


Fig. 3. Comparison of classification accuracy between our method and existing approaches on three benchmark datasets: (a) miniImageNet, (b) CIFAR-100, and (c) CUB200. The evaluation focuses on the average accuracy across all previously seen and currently introduced categories.

Table 4

Ablation study on CIFAR-100 in the 5-way 5-shot FSCIL setting.

ORTH	ETF	PAF	0	1	2	3	4	5	6	7	8	Acc	DR	FRI
			82.52	76.82	73.34	69.68	66.19	62.85	60.96	59.02	56.11	67.50	32.00	1.85
✓			80.37	75.44	73.58	70.29	66.27	61.45	59.23	56.93	54.50	66.45	32.19	1.81
	✓		82.64	77.98	74.01	71.28	67.54	62.25	59.32	57.35	54.82	67.47	33.66	1.76
		✓	78.33	75.23	72.96	70.51	66.37	63.54	61.67	59.79	55.43	67.09	29.24	2.00
	✓	✓	82.73	78.21	74.18	71.54	67.79	63.86	61.92	60.23	58.27	68.75	29.57	2.02

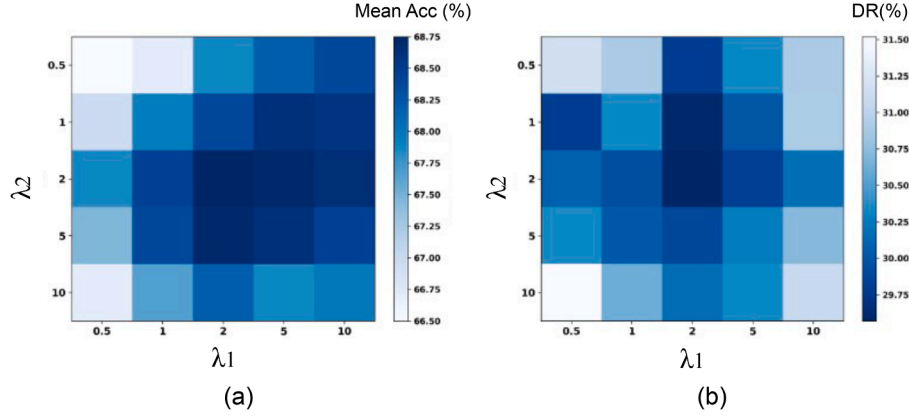


Fig. 4. Selection of λ_1 and λ_2 Values on CIFAR-100, evaluated by (a) mean accuracy and (b) degradation rate.

Although some previous methods exhibit relatively low forgetting scores, their overall classification accuracies remain significantly inferior. This suggests that the apparent stability is primarily due to limited generalization ability rather than effective knowledge retention. Therefore, we propose a comprehensive anti-forgetting performance index that combines the overall classification accuracy and the forgetting rate for each session to evaluate the anti-forgetting performance of the model method. The formula is as follows:

$$\text{FRI} = \frac{A_{\text{mean}}}{\ln(F) + \text{DR}}, \quad (24)$$

where A_{mean} is the mean accuracy across all sessions, and DR stands for performance degradation rate. This metric jointly reflects classification performance and forgetting robustness, providing a more holistic evaluation of model effectiveness in FSCIL scenarios.

4.4. Compare with SOTA methods

We evaluated our method on three benchmark datasets and compared it with state-of-the-art approaches to demonstrate its effectiveness. Fig. 3 illustrates the accuracy variation of different methods across all sessions on the three datasets. The results, summarized in Tables 1–3 report classification accuracy at each session, along with the average accuracy and the performance drop in the final session. As shown, our method consistently outperforms all baselines on CIFAR-100, miniImageNet, and CUB-200-2011, achieving average accuracy improvements of 1.25%, 2.52%, and 0.28%, respectively.

The experimental results show that our method performs better across datasets, better preserves old classes and adapts to new ones, and achieves cross-dataset generalization.

4.5. Ablation study and analysis

We design ablation experiments to evaluate the individual and combined contributions of the attention-based feature distributor and the PAF freezing strategy. We also compare our method with the NC-FSCIL baseline to analyze the performance gains introduced by each component.

Table 4 presents the ablation results on the CIFAR-100 dataset. It can be observed that replacing the orthogonal classifier with the ETF classifier improves the average accuracy by 1.02%. Similarly, using PAF alone, which selectively freezes important parameters, reduces the final performance drop by 2.76%. When combining both ETF and PAF, the model achieves the highest average accuracy of 68.75%, significantly outperforming all other configurations, especially in later sessions when catastrophic forgetting is most severe. These results confirm that the proposed designs are not only effective individually, but also complementary when integrated.

4.6. Hyperparameter analysis

To further evaluate the impact of key hyperparameters, we conducted a comparative analysis of the ETF weight λ_1 and the PAF regularization weight λ_2 . We performed experiments with different combinations of λ_1 and λ_2 within the range $\{0.5, 1, 2, 5, 10\}$, analyzing their effect on model performance. Fig. 4 shows how changes in these two parameters affect the model's mean accuracy and the rate of performance degradation on CIFAR-100. Our experimental results indicate that the combination of $\lambda_1 = 5$ and $\lambda_2 = 5$ yields the best performance.

5. Discussion

The VPAF-FSCIL method, based on feature decoupling and parameter-adaptive freezing, demonstrates significant effectiveness in improving feature stability and class adaptability, but it also presents issues and potential limitations that warrant further exploration.

First, we analyze the method's computational complexity. The algorithm achieves its highest complexity within the PAF strategy due to the need to calculate parameter importance. After the base pre-training stage, we use a second-order moment diagonal approximation to compute FIM, which represents the importance of each parameter for the task. The complexity of this computation is $O(B \times P)$, where B is the number of samples and P is the number of parameters in the network. The complexity of this method depends on the number of samples and model parameters. As model size increases, computational cost rises gradually, potentially forming a computational bottleneck.

Secondly, under the FSCIL setting, the scarcity of new-class samples makes the model prone to overfitting to new classes. Our method mitigates this issue by introducing the PAF strategy, which selectively freezes network parameters critical for old classes, thus preventing overfitting caused by the scarcity of new-class samples. Meanwhile, the feature decoupling design optimizes the allocation of new class embeddings, further stabilizing old class feature distributions and maintaining inter-class geometric separability. In the future, we will explore new strategies to improve the model's generalization ability and robustness in complex environments.

Experiments show that the proposed method achieves good results on standard benchmark datasets. However, its generalization to real-world complex scenarios may still face challenges. For instance, in autonomous driving, new object categories or dynamic scenes may violate the feature-space stability assumption; in medical diagnosis, new disease categories often exhibit domain shifts or high similarity, complicating class-space allocation and prototype alignment. Future work could integrate domain adaptation or uncertainty estimation techniques to improve robustness and reliability in practical applications.

6. Conclusion

In this paper, we proposed a novel method for FSCIL that simultaneously enhances the stability of old class representations and the adaptability of new classes. To address the challenges of representation drift and inter-class imbalance, we designed a feature-decoupled architecture consisting of a Feature Extractor, a Feature Distributor, and an ETF-based Classifier. The Feature Distributor incorporates an adaptive attention mechanism to construct data-driven virtual class space allocations, enabling smooth and discriminative embedding of novel classes. To further mitigate catastrophic forgetting, we introduced a Parameter-Adaptive Freezing strategy, which evaluates parameter importance using Fisher Information and applies a soft freezing mechanism to selectively preserve critical parameters. Additionally, we proposed a forgetting-aware evaluation metric to better capture the forgetting dynamics across incremental sessions. Extensive experiments on three benchmark datasets demonstrate that our method consistently outperforms state-of-the-art approaches in both accuracy and stability, validating the effectiveness of our contributions.

CRedit authorship contribution statement

Ye Yao: Writing – review & editing, Writing – original draft, Validation; **Junxi Li:** Writing – review & editing; **Xiong Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Akmal, F., Meng, F., Wu, Q., Chen, S., Zhang, R., & Assefa, M. (2024). Class similarity weighted knowledge distillation for few shot incremental learning. *Neurocomputing*, 584, 127587.
- Aljundi, R., Babiloni, F., Elhoseiny, M., Rohrbach, M., & Tuytelaars, T. (2018). Memory aware synapses: Learning what (not) to forget. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 139–154).
- An, Z., Sun, G., Liu, Y., Li, R., Han, J., Konukoglu, E., & Belongie, S. (2025). Generalized few-shot 3d point cloud segmentation with vision-language model. In *Proceedings of the computer vision and pattern recognition conference* (pp. 16997–17007).
- An, Z., Sun, G., Liu, Y., Li, R., Wu, M., Cheng, M.-M., Konukoglu, E., & Belongie, S. (2024a). Multimodality helps few-shot 3d point cloud semantic segmentation. *arXiv:2410.22489*.
- An, Z., Sun, G., Liu, Y., Liu, F., Wu, Z., Wang, D., Van Gool, L., & Belongie, S. (2024b). Re-thinking few-shot 3d point cloud semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3996–4006).
- Chaudhry, A., Ranzato, M., Rohrbach, M., & Elhoseiny, M. (2018). Efficient lifelong learning with a-gem. *arXiv:1812.00420*.
- Chi, Z., Gu, L., Liu, H., Wang, Y., Yu, Y., & Tang, J. (2022). Meta-fscil: A meta-learning approach for few-shot class incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14166–14175).
- Cui, Y., Xiong, W., Tavakolian, M., & Liu, L. (2021). Semi-supervised few-shot class-incremental learning. In *2021 IEEE International conference on image processing (ICIP)* (pp. 1239–1243). IEEE.
- De Lange, M., Aljundi, R., Masana, M., Parisot, S., Jia, X., Leonardis, A., Slabaugh, G., & Tuytelaars, T. (2021). A continual learning survey: Defying forgetting in classification tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(7), 3366–3385.
- Dong, S., Hong, X., Tao, X., Chang, X., Wei, X., & Gong, Y. (2021). Few-shot class-incremental learning via relation knowledge distillation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1255–1263). (vol. 35).
- Finn, C., Abbeel, P., & Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning* (pp. 1126–1135). PMLR.
- Goodfellow, I. J., Mirza, M., Xiao, D., Courville, A., & Bengio, Y. (2013). An empirical investigation of catastrophic forgetting in gradient-based neural networks. *arXiv:1312.6211*.
- Hao, M., Gu, Y., Dong, K., Tiwari, P., Lv, X., & Ning, X. (2025). A prompt regularization approach to enhance few-shot class-incremental learning with two-stage classifier. *Neural Networks*, 188, 107453.
- Hersche, M., Karunaratne, G., Cherubini, G., Benini, L., Sebastian, A., & Rahimi, A. (2022). Constrained few-shot class-incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9057–9067).
- Hsu, Y.-C., Liu, Y.-C., Ramasamy, A., & Kira, Z. (2018). Re-evaluating continual learning scenarios: A categorization and case for strong baselines. *arXiv:1810.12488*.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- Li, Z., & Hoiem, D. (2017). Learning without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(12), 2935–2947.
- Liu, H., Gu, L., Chi, Z., Wang, Y., Yu, Y., Chen, J., & Tang, J. (2022). Few-shot class-incremental learning via entropy-regularized data-free replay. In *European conference on computer vision* (pp. 146–162). Springer.
- Mallya, A., & Lazebnik, S. (2018). Packnet: Adding multiple tasks to a single network by iterative pruning. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7765–7773).
- Papayan, V., Han, X. Y., & Donoho, D. L. (2020). Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40), 24652–24663.
- Parnami, A., & Lee, M. (2022). Learning from few examples: A summary of approaches to few-shot learning. *arXiv:2203.04291*.
- Rebuffi, S.-A., Kolesnikov, A., Sperl, G., & Lampert, C. H. (2017). icarl: Incremental classifier and representation learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2001–2010).
- Ren, M., Triantafyllou, E., Ravi, S., Snell, J., Swersky, K., Tenenbaum, J. B., Larochelle, H., & Zemel, R. S. (2018). Meta-learning for semi-supervised few-shot classification. *arXiv:1803.00676*.
- Serra, J., Suris, D., Miron, M., & Karatzoglou, A. (2018). Overcoming catastrophic forgetting with hard attention to the task. In *International conference on machine learning* (pp. 4548–4557). PMLR.
- Shi, G., Chen, J., Zhang, W., Zhan, L.-M., & Wu, X.-M. (2021). Overcoming catastrophic forgetting in incremental few-shot learning by finding flat minima. *Advances in Neural Information Processing Systems*, 34, 6747–6761.
- Shin, H., Lee, J. K., Kim, J., & Kim, J. (2017). Continual learning with deep generative replay. *Advances in Neural Information Processing Systems*, 30, 2990–2999.
- Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical networks for few-shot learning. *Advances in Neural Information Processing Systems*, 30, 4077–4087.
- Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1–9).
- Tao, X., Hong, X., Chang, X., Dong, S., Wei, X., & Gong, Y. (2020). Few-shot class-incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 12183–12192).
- Tian, S., Li, L., Li, W., Ran, H., Ning, X., & Tiwari, P. (2024). A survey on few-shot class-incremental learning. *Neural Networks*, 169, 307–324.
- Wang, X., Ji, Z., Pang, Y., & Yu, Y. (2024). A cognition-driven framework for few-shot class-incremental learning. *Neurocomputing*, 600, 128118.
- Wang, Y., Chao, W.-L., Weinberger, K. Q., & Van Der Maaten, L. (2019). Simpleshot: Re-visiting nearest-neighbor classification for few-shot learning. *arXiv:1911.04623*.
- Xing, C., Rostamzadeh, N., Oreshkin, B., & Pedro O, O. P. (2019). Adaptive cross-modal few-shot learning. *Advances in Neural Information Processing Systems*, 32, 4847–4857.
- Yang, B., Lin, M., Zhang, Y., Liu, B., Liang, X., Ji, R., & Ye, Q. (2022). Dynamic support network for few-shot class incremental learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3), 2945–2951.
- Yang, Y., Yuan, H., Li, X., Lin, Z., Torr, P., & Tao, D. (2023). Neural collapse inspired feature-classifier alignment for few-shot class incremental learning. *arXiv:2302.03004*.
- Yoon, J., Yang, E., Lee, J., & Hwang, S. J. (2017). Lifelong learning with dynamically expandable networks. *arXiv:1708.01547*.
- Zhou, D.-W., Wang, F.-Y., Ye, H.-J., Ma, L., Pu, S., & Zhan, D.-C. (2022a). Forward compatible few-shot class-incremental learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9046–9056).
- Zhou, D.-W., Ye, H.-J., Ma, L., Xie, D., Pu, S., & Zhan, D.-C. (2022b). Few-shot class-incremental learning by sampling multi-phase tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(11), 12816–12831.
- Zou, Y., Zhang, S., Li, Y., & Li, R. (2022). Margin-based few-shot class-incremental learning with class-level overfitting mitigation. *Advances in Neural Information Processing Systems*, 35, 27267–27279.